

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - VII

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Hash Table:

Write a program to:

- a. Implement a hash table with separate chaining for collision handling.

CODE:

```
size = 5
table = [[]for _ in range(size)]

def insert(key,value):
    table[(key%size)].append((key,value))

insert(10,"Rahul")
insert(15,"Nitesh")
insert(20,"Raj")
insert(32,"Amit")
insert(33,"Jay")

print(table)

print("Rahul_51")
|
```

OUTPUT:

```
===== RESTART: C:/Users/IT-50/Desktop/Practical08.py =====
[[ (10, 'Rahul'), (15, 'Nitish'), (20, 'Raj') ], [], [(32, 'Amit')], [(33, 'Jay')], []]
Rahul_51
>> |
```

- b. Store and retrieve data from the hash table.

CODE:

```
size = 5
table = [[]for _ in range(size)]

def insert(key,value):
    table[(key%size)].append((key,value))

def search(key):
    for k,v in table[key%size]:
        if k==key:
            return v
    return "Not Found"

insert(10,"Rahul")
insert(15,"Nitesh")
insert(20,"Raj")
insert(32,"Amit")
insert(33,"Jay")

print(table)

key = int(input("Enter key : "))
print(search(key))

print("Rahul_51")
```

OUTPUT:

```
===== RESTART: C:/Users/IT-50/Desktop/Practical08.py =====
[[ (10, 'Rahul'), (15, 'Nitesh'), (20, 'Raj') ], [], [(32, 'Amit')], [(33, 'Jay')], []]
Enter key : 10
Rahul
Rahul_51

===== RESTART: C:/Users/IT-50/Desktop/Practical08.py =====
[[ (10, 'Rahul'), (15, 'Nitesh'), (20, 'Raj') ], [], [(32, 'Amit')], [(33, 'Jay')], []]
Enter key : 15
Nitesh
Rahul_51

===== RESTART: C:/Users/IT-50/Desktop/Practical08.py =====
[[ (10, 'Rahul'), (15, 'Nitesh'), (20, 'Raj') ], [], [(32, 'Amit')], [(33, 'Jay')], []]
Enter key : 33
Jay
Rahul_51

===== RESTART: C:/Users/IT-50/Desktop/Practical08.py =====
[[ (10, 'Rahul'), (15, 'Nitesh'), (20, 'Raj') ], [], [(32, 'Amit')], [(33, 'Jay')], []]
Enter key : 100
Not Found
Rahul_51
```